FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IX

JEE (Main)-2021

TEST DATE: 04-04-2021

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

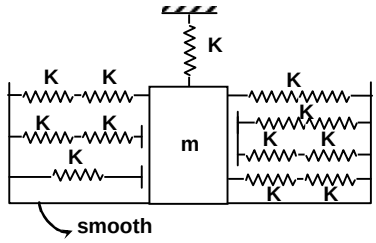
PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A resonant pipe closed at one end is marked 'C' and another similar pipe open at both ends is marked 'O'. then,
   (A) the resonant frequencies in the fundamental mode are equal when the length of pipe C is half the length of pipe O.
   (B) the resonant frequencies in the fundamental mode are equal when the length of pipe O is half the length of pipe C.
   (C) when the pipes are of equal length, their first overtone frequencies are same.
   (D) none of these

2. The time period of small horizontal oscillation of the given block is

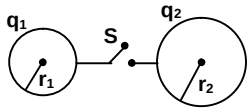


   (A) $T = 2\pi\sqrt{\frac{m}{7K}}$
   (B) $T = 2\pi\sqrt{\frac{2m}{7K}}$
   (C) $T = 2\pi\sqrt{\frac{2m}{5K}}$
   (D) $T = 4\pi\sqrt{\frac{m}{7K}}$

3. If $Q = \frac{x^n}{y^m}$ and Δx, Δy are absolute errors in the measurement of x and y, then absolute error ΔQ in Q is
   (A) $\Delta Q = \pm\left(n\frac{\Delta x}{x} + m\frac{\Delta y}{y}\right)$
   (B) $\Delta Q = \pm\left(n\frac{\Delta x}{x} + m\frac{\Delta y}{y}\right)Q$
   (C) $\Delta Q = \pm\left(n\frac{\Delta x}{x} - m\frac{\Delta y}{y}\right)Q$
   (D) $\Delta Q = \pm\left(n\frac{\Delta x}{y} - m\frac{\Delta y}{x}\right)Q$

4. In a series LCR circuit, the frequencies f_1 and f_2 at which the current amplitude falls to $1/\sqrt{2}$ of the current amplitude at resonance are separated by frequency interval
   (A) $\frac{R}{2\pi L}$
   (B) $2\pi\sqrt{LC}$
   (C) $\sqrt{LC}$
   (D) RC

5. A system consists of two metallic spheres of radii r_1 and r_2 connected by a thin wire and a switch S as shown. Initially S is open and spheres carry charges q_1 and q_2 respectively. If the switch 'S' is closed, the potential of the system is



(A) $\left(\frac{1}{4\pi\varepsilon_0}\right)\left\{\frac{q_1q_2}{r_1r_2}\right\}$

(B) $\left(\frac{1}{4\pi\varepsilon_0}\right)\left\{\frac{(q_1+q_2)}{(r_1+r_2)}\right\}$

(C) $\left(\frac{1}{4\pi\varepsilon_0}\right)\left\{\frac{q_1}{r_1}+\frac{q_2}{r_2}\right\}$

(D) $\left(\frac{1}{4\pi\varepsilon_0}\right)\left\{\frac{(q_1+q_2)}{(r_1r_2)^{1/2}}\right\}$

6. A nuclear reactor containing U^{235} produces 10 MW. Calculate the fission rate assuming that 200 MeV of useful energy is released in one fission?

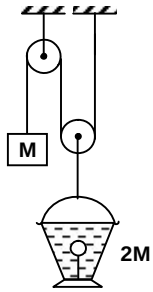
(A) 3.1×10^{17} sec^{-1}

(B) 6×10^{8} sec^{-1}

(C) 1.5×10^{17} sec^{-1}

(D) 2×10^{17} sec^{-1}

7. Consider the system shown in the figure. A ball of density σ is tied to the bottom of the bucket which contains a liquid of density ρ ($\rho>\sigma$). The volume of the ball is V and the mass of bucket along with its contents is 2M. The system is released from rest. The thread connected to the ball is taut. The tension in the thread connected to the ball will be:



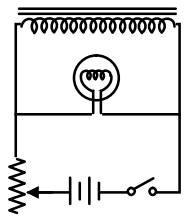
(A) greater than $(\rho-\sigma)Vg$

(B) less than $(\rho-\sigma)Vg$

(C) equal to $(\rho-\sigma)Vg$

(D) none of these

8. For the circuit arrangement shown in the figure, the circuit is switched on and the variable resistance is adjusted such that the bulb just glows. Now if the circuit is switched off



(A) the bulb stops glowing

(B) momentarily the bulb becomes dim and then stops glowing.

(C) momentarily the bulb becomes very bright and then stops glowing

(D) the bulb continues to glow for some time and then stops glowing.

9. There is a string of length 3L whose end points are fixed. The velocity of transverse wave in the string is v. It is desired that the mid point of the string should be a node. The minimum excitation frequency to achieve this will be

(A) 2v/3L

(B) v/6L

(C) v/2L

(D) v/3L

10. Two very long, straight, parallel wires carry steady currents 'I' and '–I' respectively. The distance between the wires is 'd'. At a certain instant of time a point charge 'q' is at a point equidistant from the two wires, in the plane of the wires. Its instantaneous velocity 'v' is perpendicular to the plane of wires. The magnitude of the force due to the magnetic field acting on the charge at this instant is

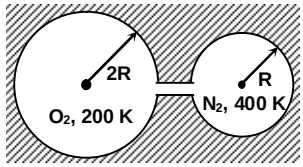
(A) $\mu_0 Iqv/(2\pi d)$

(B) $2\mu_0 Iqv/(\pi d)$
(C) $\mu_0 Iqv/(\pi d)$
(D) zero

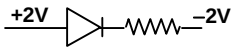
11. Which of the following combinations of three different physical quantities P, Q and R can never be a meaningful physical quantity?
(A) PQ–R
(B) PQ/R
(C) (P–Q)/R
(D) $(PR - Q^2)/QR$

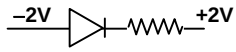
12. Two identical rings each of radius R are placed coaxially such that the distance between their centre is H. Now a soap solution film is formed between the two rings such that there is no films in the circular regions bounded by the rings. Let T be the surface tension of the solution then the total surface energy of the film will be:
(A) $4\pi RHT$
(B) more than $4\pi RHT$
(C) Less than $4\pi RHT$
(D) $2\pi RHT$

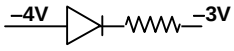
13. A thermally insulated vessel has two spherical chambers separated by a stop cock. The left part contains 1 mole of oxygen (O_2) at 200 K while the right part contains 1 mole of nitrogen (N_2) at 400 K. If the stopcock is opened, the final temperature upon the mixing of gases will be:

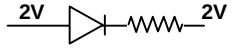


(A) between 295 and 299 K
(B) between 301 and 310 K
(C) between 199 and 301 K
(D) none of these

14. The forward biased diode is
(A) 

(B) 

(C) 

(D) 

15. Standing waves given by $y = a \sin(kx) \cos(\omega t)$ are set up in a homogeneous rod with cross sectional area S and density ρ. What is the total mechanical energy stored between the two adjacent nodes?
(A) $\frac{\rho a^2 \omega^2 S\pi}{8K}$
(B) $\frac{\rho a^2 \omega^2 S\pi}{2K}$
(C) $\frac{\rho a^2 \omega^2 S\pi}{4K}$
(D) $\frac{\rho a^2 \omega^2 S\pi}{4K} \sin^2 \omega t$

16. When light is incident from air to glass at an angle 57°, the reflected beam is completely polarised. If the same beam is incident from water to glass, the angle of incidence at which reflected beam is completely polarised will be
(A) $\theta = 57°$
(B) $\theta > 57°$

(C) $\theta < 57°$
(D) cannot be determined

17. A thin rod is bent in the shape of a small circle of radius 'r'. If the charge per unit length of the rod is σ, and if the circle is rotated about its axis at a rate of 'n' rotations per second, the magnetic induction at a point on the axis at a large distance y from the centre is

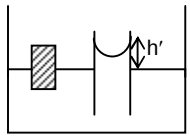
(A) $\frac{\mu_0 \pi r^3 n\sigma}{y^3}$

(B) $\frac{2\mu_0 \pi r^3 n\sigma}{y^3}$

(C) $\frac{\mu_0 r^3 n\sigma}{4\pi y^3}$

(D) $\frac{\mu_0 r^3 n\sigma}{2\pi y^3}$

18. In a capillary tube placed inside the liquid of density (ρ) in a container, the rise of liquid is h. When a block of density 'σ' is placed on the liquid as shown in the figure, the rise of liquid in the tube is h′. If $\sigma < \rho$ then



(A) $h' = h$
(B) $h' < h$
(C) $h' > h$
(D) insufficient data

19. A certain spherical object immersed in a liquid of viscosity η is moved along a straight line according to the equation $x = A \sin(\omega t)$ where x is the x coordinate of its centre. The radius of the object is R. What is the net work done by viscous forces on the body between $t = 0$ and $t = 2\pi/\omega$?

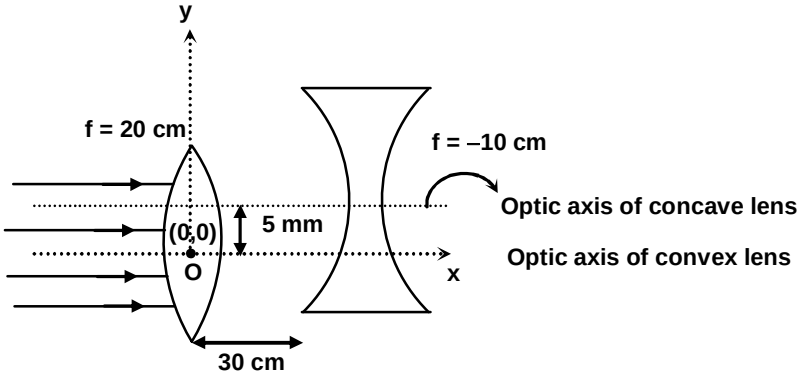
(A) $\frac{2}{9}\pi^2 \eta R A^2 \omega$

(B) $-\frac{2}{9}\pi^2 \eta R A^2 \omega$

(C) $-6\pi^2 \eta R A^2 \omega$

(D) $6\pi^2 \eta R A^2 \omega$

20. If the optic axes of convex and concave lenses are separated by a distance 5 mm as shown in the figure. Find the co-ordinates of the final image formed by the combination of lenses if a parallel beam of light is incident on the convex lens. Origin is taken at the optical centre of the convex lens and x-axis along its optic axis.



(A) (25 cm, 0.5 cm)

(B) (25 cm, 0.25 cm)
(C) (25 cm, 0.5 cm)
(D) (25 cm, -0.25 cm)

SECTION – B
(Numerical Answer Type)

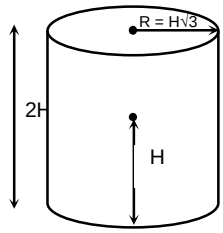
This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. Liquid A is at a temperature of 0°C while liquid B is at a temperature of 9°C. If half of liquid B were to be mixed with liquid A, the temperature would be 4.5°C. Let the final temperature in Celsius if entire liquid B were to be mixed with entire liquid A is T. Find the value of $\frac{T}{10}$.

22. A stationary observer O is located at some distance from site of a bomb blast such that if no wind were to blow the sound of the blast would have reached his ears in $5\sqrt{3}$ seconds. However since a wind was blowing perpendicular to the line joining the observer to the site of blast, the observer heard the sound of blast after 10 seconds. How many times is the speed of wind as compared to the speed of sound in still air.

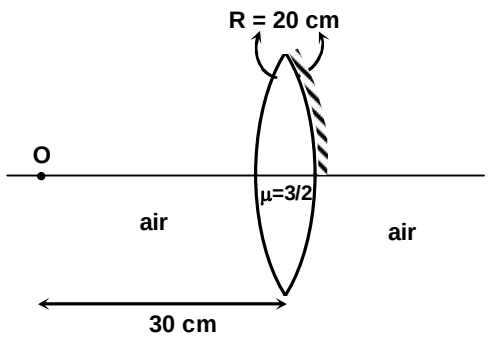
23. A certain satellite revolving around the earth in a circular orbit of radius R, takes 4 days to complete one full revolution. If the same satellite were to be put in an orbit of radius R/2 around another planet, it will take 2 days to complete one full revolution. The ratio of mass of the earth to the mass of the other planet is x. Find the value of $\left(\frac{x}{8}\right)$.

24. A small spherical black body having surface area 'A' and absolute temperature T is kept at the geometric centre of a cylindrical can of height 2H and radius $H\sqrt{3}$ as shown in the figure. The lid of the can is missing. The portion of power which is radiated by the black body and escapes directly from the surface where the lid is missing is given by $P_{Lid} = \frac{\sigma AT^4}{\eta}$. Then find the value of $\frac{\eta}{8}$.



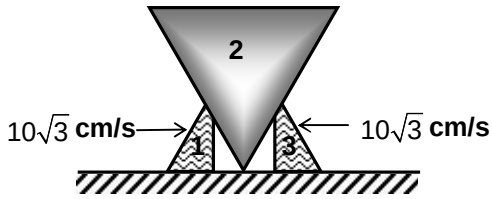
25. Electrons in a hydrogen like atom (Z = 3) make transitions from the fourth excited state to the third excited state and from the third excited state to the second excited state. The resulting radiations are incident on a metal plate to eject photoelectrons. The stopping potential for photoelectrons ejected by shorter wavelength is 3.95 V. If the stopping potential for the photoelectrons ejected by the longer wavelength is N volt, then find N. (Take energy of the hydrogen atom in ground state as -13.6 eV)

26. A point object O is placed at a distance 30 cm from a thin equiconvex lens (μ = 3/2) on its optic axis as shown in the figure. If the distance between the two images formed by the upper and the lower part of the lens is x meter. Find the value of x.

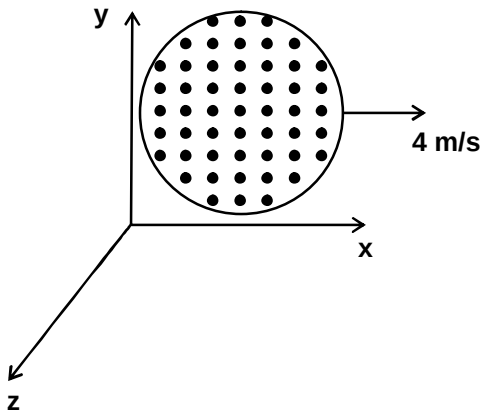


27. A point source of power 50 π watts is producing sound waves. The velocity of sound is 330 m/s, density of air is 1.0 kgm^{-3}. Then (using π = 22/7) at r = $\sqrt{330}$ m from the point source, the pressure amplitude is x Nm^{-2}. Find the value of $\frac{x}{2}$.

28. One equilateral wedge and two other wedges are arranged on a rough horizontal surface as shown in the figure. Now wedge 1 and 3 are moved horizontally with the velocity $10\sqrt{3}$ cm/s as shown. If velocity of the wedge-2 is x cm/s. Find the value of x.



29. A uniform magnetic field $\vec{B} = 0.25(\hat{k})$ T exists in a circular region of radius R = 5m. A conducting loop of radius R = 5m lying in the x-y plane encloses the magnetic field at t = 0 and then pulled with a uniform velocity $\vec{v} = 4\hat{i}$ m/s. If the emf induced in the loop at t = 2 sec is E volt. Find the value of $\left(\frac{E}{24}\right)$.



30. A thin conducting wire in the shape of a 'figure of eight' is situated with its circular loops in two planes making an angle of 120° with each other if the current in the loop is I and the radius is R, the magnetic induction at a point of intersection P of the two axes of the loops is $\left(\frac{n\mu_0 i}{8R}\right)$. Find the value of n.

Chemistry

PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. The number of hydrogen atoms in 0.9 g of galactose is same as
    (A) 0.048 g of hydrazine
    (B) 0.17 of ammonia
    (C) 0.30 g of ethane
    (D) 0.03 g of hydrogen

32. Which of the following statement is INCORRECT for Lucas Test which is used to distinguish between $1^{o}, 2^{o}$ and 3° alcohols.
    (A) Allyl alcohol and benzyl alcohol give immediate turbidity.
    (B) Alcohols which reacts readily with Na-metals will give turbidity at a much faster rate.
    (C) Alcohols which gives red colour during Victor Mayer test, will give turbidity at a much slower rate.
    (D) Lucas test involves Acid-Base reaction.

33. In presence of PbS abd ZnS in a mineral, the froth floatation process requires NaCN because
    (A) ZnS forms a soluble complex $\left[Zn(CN)_4\right]^{--}$.
    (B) $Zn(CN)_2$ precipitated.
    (C) PbS forms a soluble complex $Na_2\left[Pb(CN)_4\right]$.
    (D) ZnS forms an insoluble complex $\left[Zn(CN)_4\right]^{2-}$.

34. Which of the following statement is INCORRECT for 'Electrical Disintegration' method for preparation of colloids?
    (A) It is also known as Bredig's arc method.
    (B) It involves both vaporization and condensation.
    (C) It is used to prepare gold, silver and platinum sols.
    (D) It needs an electric arc between the electrodes of the metal immersed in the dispersed phase.

35. Which of the following antibiotic is bactericidal in nature
    (A) Erythromycin
    (B) Ofloxacin
    (C) Tetracycline
    (D) Chloramphenicol

36. Which of the following amino acid consists of two $-NH_2$ group in it
    (A) Isoleucine
    (B) Aspartic acid
    (C) Glutamic acid
    (D) Lysine

37. Which of the following statement regarding melamine is INCORRECT?
    (A) It is aromatic
    (B) It contains 6 pairs of non-bonding electrons.

(C) Its molecular formula is $C_3H_8N_6$.

(D) it is a heterocyclic compound.

38. A metal crystallizes in bcc lattice. The % of edge length not covered by atom is

(A) 10.4%

(B) 13.4%

(C) 12.5%

(D) 16%

39. Both temporary and permanent hardness is removed on boiling water with

(A) $Ca(OH)_2$

(B) $NaHCO_3$

(C) Na_2CO_3

(D) CaO

40. The increasing order of solubility of BaS, MnS, Sb_2S_3 and K_2S is

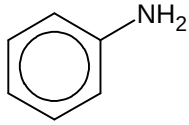
(A) $BaS < MnS < Sb_2S_3 < K_2S$

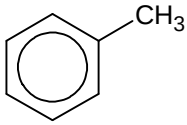
(B) $K_2S < MnS < Sb_2S_3 < BaS$

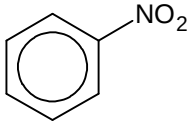
(C) $Sb_2S_3 < MnS < BaS < K_2S$

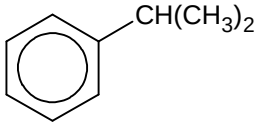
(D) $Sb_2S_3 < BaS < MnS < K_2S$

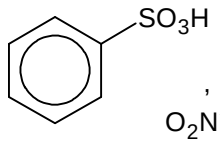
41. Which of the following compounds will undergo Friedel-Craft alkylation with fastest rate?

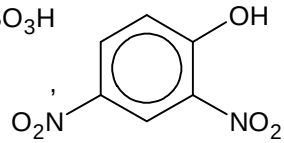
(A) 

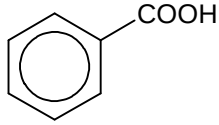
(B) 

(C) 

(D) 

42. When 

, 

and 

are treated with $NaHCO_3$ solution, the gases released are

(A) SO_2, NO_2, CO_2

(B) SO_3, N_2O, CO_2

(C) CO_2, CO_2, CO_2

(D) CO_2, NO_2, CO_2

43. Select the correct statement about van der Waals constants 'a' and 'b'
(A) a, b are constants for a gas at any temp. and pressure.
(B) a, b are not strictly constants and vary with temperature and pressure.
(C) The values of a, b are same according to different equations for real gases.
(D) For real gases, $T_B = \frac{2a}{Rb}$ where T_B is the Boyle's temperature.

44. Consider two liquids (A) (B) having vapour pressures P_A^o and P_B^o in their pure forms respectively. The plot of $\frac{1}{x_A}$ vs $\frac{1}{y_A}$ (where x_A and y_A are the mole fractions of liquid (A) in liquid and vapour phase respectively, is linear with slope and y-intercept respectively
(A) $\frac{P_A^o}{P_B^o}, \frac{P_A^o - P_B^o}{P_B^o}$
(B) $\frac{P_A^o}{P_B^o}, \frac{P_B^o - P_A^o}{P_B^o}$
(C) $\frac{P_B^o}{P_A^o}, \frac{P_A^o - P_B^o}{P_B^o}$
(D) $\frac{P_B^o}{P_A^o}, \frac{P_B^o - P_A^o}{P_B^o}$

45. Which of the following is not obtained as a product during complete hydrolysis of XeF_4?
(A) XeO_3
(B) Xe
(C) HF
(D) XeO_2F_2

46. Amphoteric oxides of Mn and Cr are
(A) MnO, CrO
(B) Mn_2O_3, Cr_2O_3
(C) MnO_2, Cr_2O_3
(D) Mn_2O_7, CrO_3

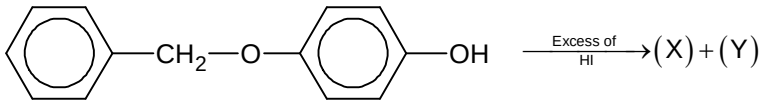
47. The complex having the highest magnetic moment is
(A) $\left[Zn(H_2O)_6\right]^{++}$
(B) $\left[Cr(H_2O)_6\right]^{++}$
(C) $\left[Mn(H_2O)_6\right]^{++}$
(D) $\left[Cu(H_2O)_6\right]^{++}$

48. Consider the following reactions
$(CH_3)_3C-Cl \xrightarrow{LiAlH_4} (X)$
$CH_3CH=CHCH_2-Cl \xrightarrow{Ph_3SnH} (Y)$
(X) and (Y) respectively are:
(A) $(CH_3)_3CH,\ CH_3(CH_2)_2CH_3$
(B) $CH_2=C(CH_3)_2,\ CH_3CH=CHCH_3$

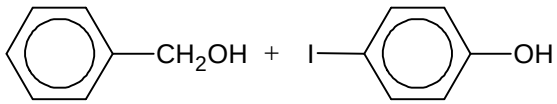
(C) $CH_2 = C(CH_3)_2$, $CH_3CH_2CH_2CH_2Cl$

(D) $(CH_3)_3CH$, $CH_3CH_2CH_2CH_2Cl$

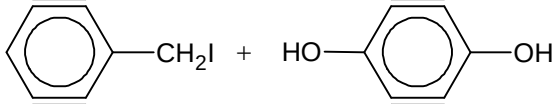
49.



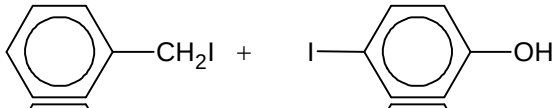
(A)



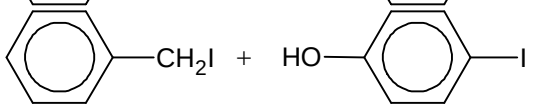
(B)



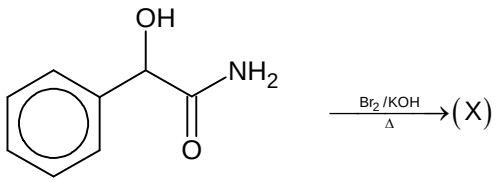
(C)



(D)

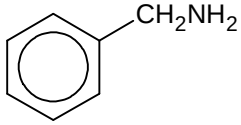


50.

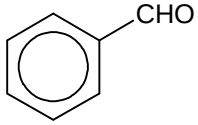


(X) is:

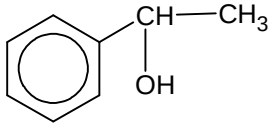
(A)



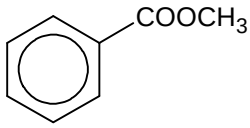
(B)



(C)



(D)



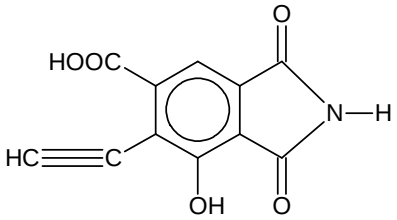
SECTION – B
(Numerical Answer Type)

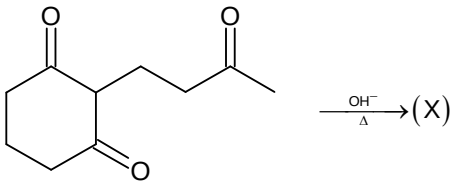
This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. For the reaction $X(g) \longrightarrow Y(g) + \frac{1}{2}Z(g)$, the pressure increases from 150 mm Hg to 170 mm Hg in 10 min. The rate of disappearance of X in mm per min. is…..

52. Electrolysis of hot aqueous solution of NaCl gives $NaClO_4$. Then the number of Faraday required to deposit 500 g of $NaClO_4$ is….
(Molar mass of Na = 23 g mol^{-1}, Cl = 35.5 g mol^{-1}, O = 16 g mol^{-1})

53. How many mols of NaOH are required for complete neutralization of the compound



54. 

The degree of unsaturation of (X) is….

55. The number of chlorine atom(s) which is/are directly bonded to each beryllium in solid beryllium chloride is….

56. A triatomic non linear gas is expanded according to $PV^2 = \text{Constant}$ under very high temperature. What is the molar heat capacity of the gas in cal/mol $\left(\text{Take } R = 2 \text{ cal mol}^{-1}K^{-1}\right)$.

57. If x is the number of exchanges possible for $3d^4$ electronic configuration, then the value of $\frac{x}{4}$ is ….

58. Consider the reaction
$XY(g) \rightleftharpoons X(g) + Y(g)$
If at a given temperature $\frac{1}{3}$rd of XY(g) is dissociated, then the ratio of equilibrium constant (K_P) and total pressure (P) at equilibrium is $x \times 10^{-2}$, then the value of x is….

59. K_{SP} of AgCl is 1×10^{-10}, then its solubility in 0.1 M KNO_3 is 'S' mol L^{-1}, then the value of – logS is….

60. If 'x' is the number of lp – bp repulsions at 90° in I_3^- and 'y' is the number of lp – bp repulsions at 90° in ICl_4^-, then the value of $\frac{x+y}{4}$ is……

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Number of solution of the equation $|x^2 - 9| + |x^2 - 5| = 2$ is/are
(A) 1
(B) 2
(C) 4
(D) no solution

62. Number of solution of the equation $[x] + \left[x + \frac{1}{2}\right] = [2x]$ is/are (where [.] denotes greatest integer function)
(A) 1
(B) 4
(C) 8
(D) infinite solution

63. Locus of z if $|z - 2i| + |z + 4i| = 10$, (where $\left(z = x + iy; i = \sqrt{-1}\right)$) is
(A) circle
(B) parabola
(C) ellipse
(D) hyperbola

64. Equation of tangent at origin to the curve $x^3 + 2y^3 + x^2 - 4y^2 + 2x - y = 0$ is
(A) $y = 2x$
(B) $y = 3x$
(C) $y = 7x$
(D) $y = -x$

65. Coefficient of $a^{10}b^7c^3$ in $(ab + bc + ca - 2)^{10}$ is
(A) 120
(B) 140
(C) 260
(D) 720

66. $\left[\sqrt{3}\,\text{cosec}(20°) - \sec(20°)\right]$ is equal to (where [.] denotes greatest integer function)
(A) 1
(B) 2
(C) 3
(D) 4

67. Range of the function $f(x) = \sin\left(\ln\left(\frac{\sqrt{4-x^2}}{1-x}\right)\right) + \cos\left(\ln\left(\frac{\sqrt{4-x^2}}{1-x}\right)\right)$ is
(A) $[-1, 1]$
(B) $\left[-\sqrt{2}, \sqrt{2}\right]$
(C) $\left[-\sqrt{2}, 1\right]$
(D) none of these

68. Number of points where $f(x) = [x] \sin^2(\pi x)$ is not differentiable if $x \in (-7, 10)$ is/are (where [.] denotes greatest integer function)
(A) 15
(B) 17
(C) 6
(D) 0

69. A purse contains 4 copper coins and 3 sliver coins. A second purse contains 6 copper coins and 4 silver coins. A purse is chosen randomly and a coin is taken out of it, then the probability that it is a copper coin
(A) $\frac{41}{70}$
(B) $\frac{27}{70}$
(C) $\frac{1}{3}$
(D) $\frac{31}{70}$

70. $\int \frac{x^2 + x - 5}{(x-1)^2} e^x dx$ is
(A) $e^x + \frac{3e^x}{x-1} + C$
(B) $e^x + \frac{2e^x}{2x-3} + C$
(C) $e^x + \frac{e^x}{x-2} + C$
(D) none of these

71. If $|z_1| = 1$, $|z_2 - 2| = 3$, $|z_3 - 5| = 6$, then maximum value of $|2z_1 - 3z_2 - 4z_3|$ is
(A) 64
(B) 63
(C) 62
(D) 61

72. Number of non-negative integral solutions of the equation $x_1 + x_2 + x_3 + 4x_4 = 20$ is
(A) 526
(B) 536
(C) 1120
(D) 1236

73. $\sum_{n=1}^{\infty} \frac{n^2 - 3n + 1}{\underline{|n}}$ is equal to
(A) –1
(B) 1
(C) 2
(D) $\frac{5}{3}$

74. 3 positive real numbers a, b, c are such that $a^2 + 5b^2 + 4c^2 - 4ab - 4bc = 0$, then a : b : c is
(A) 1 : 2 : 3
(B) 2 : 3 : 4
(C) 1 : 3 : 4
(D) 4 : 2 : 1

75. In triangle ABC, $\frac{a}{b} = 2 + \sqrt{3}$, $\angle c = 60°$, then angle A and angle B can be
(A) 105°, 15°
(B) 100°, 20°
(C) 70°, 50°
(D) none of these

76. Let x(x + y) = 4 be the hyperbola, then asymptotes of the hyperbola are
(A) x = 0, x + y = 0
(B) x = 0, x – y = 0
(C) x = 0, y = 0
(D) x + y = 0, x – y = 0

77. The equation of the plane which is equidistant from lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x-2}{3} = \frac{y-3}{1} = \frac{z-1}{2}$ is
(A) x + 7y – 5z = 9
(B) x + 3y – 2z = 9
(C) x + 19y – 5z = 11
(D) 3x + 2y – 3z = 10

78. Let $f(x) = \left|\sin^{-1}(\sin x)\right| - \left(\frac{\pi - x}{2}\right)$, then number of solution of equation f(x) = 0 in $x \in [-\pi, \pi]$ is
(A) 1
(B) 2
(C) 3
(D) 4

79. Negation of the statement $p \to (q \wedge r)$ is
(A) $\sim p \to \sim(q \wedge r)$
(B) $\sim p \vee (q \wedge r)$
(C) $(q \wedge r) \to p$
(D) $p \wedge (\sim q \vee \sim r)$

80. If $\sum_{i=1}^{18}(x_i - 8) = 9$ and $\sum_{i=1}^{18}(x_i - 8)^2 = 45$, then standard deviation of $x_1, x_2, ... x_{18}$ is
(A) $\frac{5}{12}$
(B) $\frac{3}{2}$
(C) 5
(D) 2

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. If $\lim\limits_{x\to\infty}\left\{\left(x^6+x^5\right)^{\frac{1}{6}}-\left(x^6-x^5\right)^{\frac{1}{6}}\right\}=\ell$, then 3ℓ is equal to

82. Let $A=\begin{pmatrix}1 & 0\\ 2 & 1\end{pmatrix}$, then $A^{2021}=\begin{pmatrix}a & b\\ c & d\end{pmatrix}$, then $\frac{b+c}{a+d}$ is equal to

83. Let A be a 2×3 matrix whereas B be a 3×2 matrix, if |AB| = 2, then |BA| is equal to

84. Let $2xydy = (x^2 + y^2 + 1)dx$, f(1) = 0, then $f^2(2)$ is equal to

85. Number of points of local extremum in f(x) = ||x – 6| – |x – 8|| – 3|x| is/are

86. Number of real solution of the equation $\frac{x^3}{3}-\frac{3x^2}{2}+2x+6=0$ is/are

87. If $\bar{A}=\hat{i}-3\hat{j}+4\hat{k}$, $\bar{B}=6\hat{i}+4\hat{j}-8\hat{k}$, $\bar{C}=5\hat{i}+2\hat{j}+5\hat{k}$ and a vector $\bar{R}$ satisfies $\bar{R}\times\bar{B}=\bar{C}\times\bar{B}$, $\bar{R}\cdot\bar{A}=0$, then $\frac{|\bar{R}-\bar{C}|}{|\bar{B}|}$ is equal to

88. If $\left[\bar{a}+\bar{b}\quad \bar{b}+\bar{c}\quad \bar{c}+\bar{a}\right]=\lambda\left[\bar{a}\ \bar{b}\ \bar{c}\right]$, then λ is

89. If $\int\limits_0^{\pi/2}\sin^5 x\cos^4 x\,dx = k$, then 315k is equal to

90. If points (3, –2, –1), (2, 3, –4), (–1, 1, 2) and (4, 5, k) are coplanar, then |k| is equal to